

Automated EDA Web Interface

An AI-Assisted Streamlit Tool for End-to-End Data Analysis

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Background & Motivation

- Data Analysts routinely spend 60–80% of project time on data cleaning, EDA, and reporting.
- Repetitive tasks include handling missing values, outliers, type conversion, and generating standard plots.
- Junior analysts often struggle to decide which checks and visuals to run for a new dataset.
- Goal: build an interactive web app that automates these steps and explains the dataset using an LLM, while still allowing expert control.

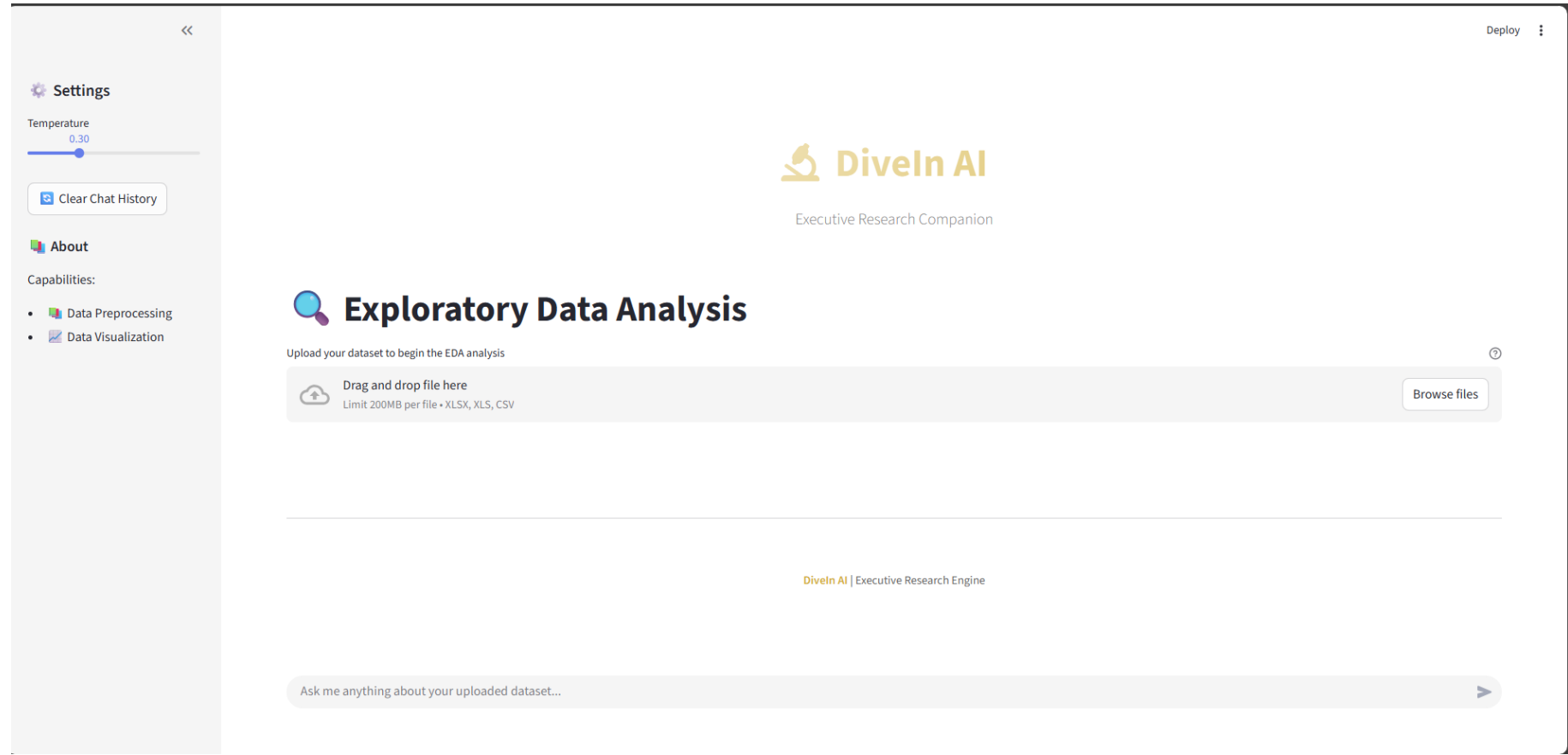
Data Source & Inputs

- Accepts user-uploaded CSV or Excel files.
- Automatically detects header row and reads into a pandas DataFrame.
- Supports heterogeneous schemas with numeric, categorical, and date-like columns.

System Architecture

- Frontend:** Streamlit multi-tab web interface.
- Backend:** pandas, NumPy, Plotly, seaborn, scikit-learn for statistics and visualization.
- LLM layer:** Google Gemini for narrative explanations and modeling suggestions.
- State management:** `st.session_state` stores original and cleaned data for before/after comparisons.
- Reporting:** ExcelWriter and ReportLab for multi-sheet Excel and PDF reports (including tables and heatmaps).

Application Interface



Streamlit interface with tabs for EDA analysis.

Key Features

1. Dataset Overview (LLM-Driven)

- Shows dataset shape, column names, data types, and memory usage.
- Displays the first 5 rows as a quick sanity check.
- An LLM prompt uses schema and sample rows to generate a 3–6 paragraph executive-level overview:
 - What the dataset represents and potential use cases.
 - Types of variables (numeric vs categorical) and key fields.
 - High-level data-quality observations (missingness, skew, potential outliers).

2. Automatic Preprocessing

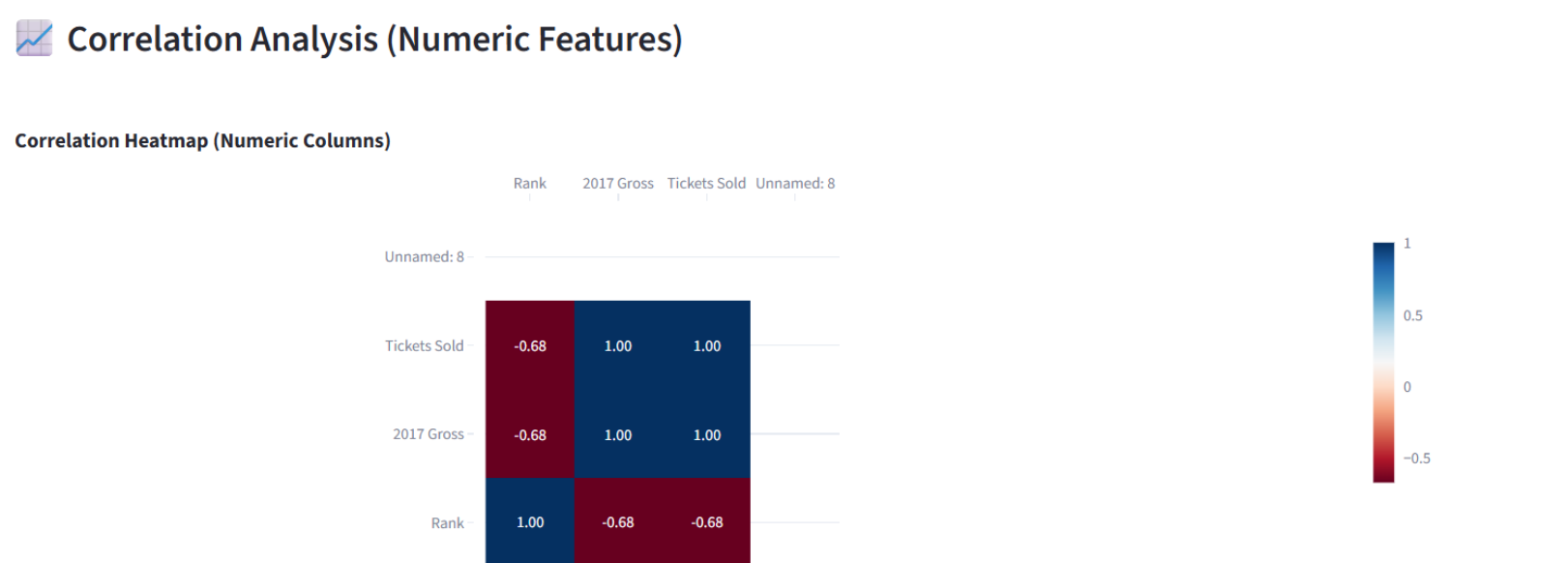
- Splits columns into numeric and categorical for tailored treatment.
- Missing values:**
 - Column-wise counts and percentages, plus “no missing values” message when clean.
 - Multiple imputation options: mean, median, mode, forward fill, backward fill.
 - Structural options: drop rows with missing values or drop columns with >50% missing.
- Outliers:**
 - IQR rule ($1.5 \times \text{IQR}$ outside Q1–Q3) applied to each numeric feature.
 - Outlier summary: column, count, percentage, lower bound, upper bound.
 - Option to cap values at the calculated bounds and store the cleaned dataset.
- Maintains both original and cleaned DataFrames for later comparison in plots and reports.

3. Advanced Analysis

- Missing-value bar charts and tables with detailed counts and percentages, before and after cleaning.
- Outlier detection table with clear numeric thresholds for each feature.
- Interactive visualizations:
 - Histograms, boxplots, and violin plots for numeric columns.
 - Bar charts and pie charts for categorical columns, focusing on top categories.
- “Before vs After” boxplots to show the effect of outlier capping and imputation.

Correlation Analysis

- Computes a Pearson correlation matrix for all numeric features.
- Displays an interactive Plotly heatmap with annotated coefficients and hover tooltips.

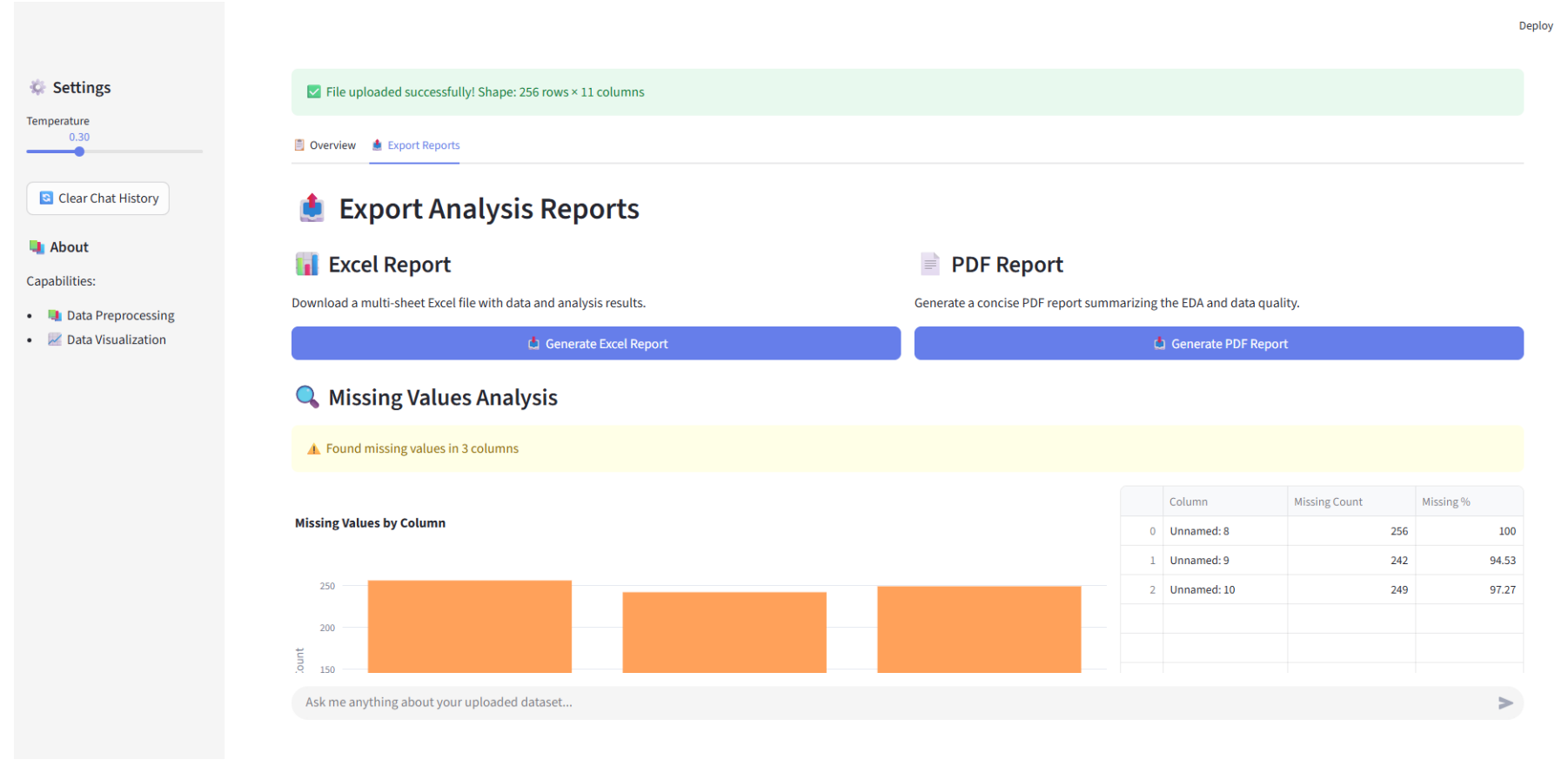


Example correlation heatmap for numeric salary features.

LLM-Powered Assistant

- Chat interface grounded on the uploaded dataframe and its statistics.
- Handles open-ended queries such as:
 - Explain the key drivers of salary in this dataset.
 - Build a regression model to predict and outline the steps.
 - Which features would you use for a classification task and why?
- Prompt design encourages the model to:
 - Summarize distributions, missing values, and outliers discovered in the app.

- Recommend ML algorithms and preprocessing pipelines (scaling, encoding, splitting).
- Provide example scikit-learn code snippets for quick model prototyping.



Reporting and Export

- Excel report:**
 - Sheets for original data, cleaned data, summary statistics, missing values, outliers, and correlation matrix.
- PDF report:**
 - Automatically generated with title page, dataset overview, tables, and correlation heatmap.
 - Designed for sharing with managers and non-technical stakeholders.

Export Analysis Reports

PDF Report

Generate a concise PDF report summarizing the EDA and data quality.

Generate PDF Report

Results & Example Insights

- Automatic detection of outliers using IQR thresholds with visual confirmation through boxplots.
- Significant reduction in manual time spent on repeated EDA steps across different projects.
- Consistent, reproducible pipeline from raw file upload to cleaned data and ready-to-use reports.

Future Work & Takeaways

- Add automated feature engineering and baseline model training, with evaluation dashboards.
- Extend the framework to time-series, text, and image datasets.
- Improve prompt-engineering and grounding for more robust dataset-aware LLM reasoning.
- Combining Streamlit, classical EDA, and LLMs produces a powerful self-service analytics tool that lowers the barrier for non-experts while remaining useful for data scientists.